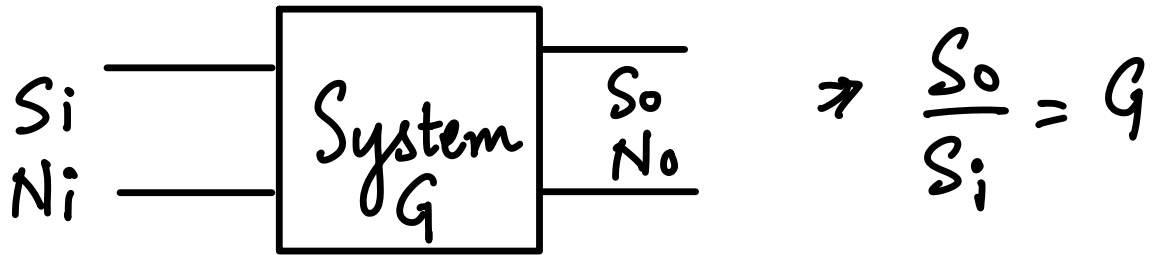




Noise Figure

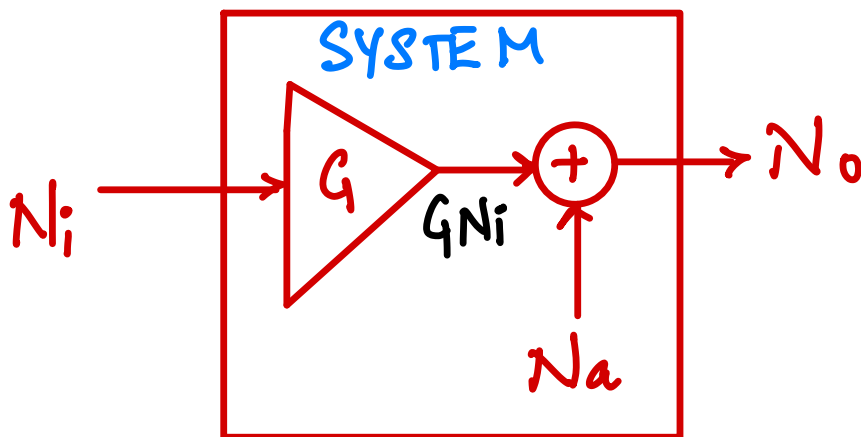


$$SNR_{i/o} = \frac{S_{i/o}}{N_{i/o}}$$

$$\text{Noise Factor } F \triangleq \frac{SNR_i}{SNR_o} = \frac{S_i/N_i}{S_o/N_o}$$

$$\text{Noise Figure} = 10 \log_{10} F$$

$$F = \frac{S_i}{S_o} \cdot \frac{N_o}{N_i} = \frac{N_o}{G N_i}$$



$$N_i = k T_o B$$

$\hookrightarrow 290K$

Equivalent Noise Temperature

$$T_e = \frac{N_a}{G} \left(\frac{1}{k B} \right) \quad G \underbrace{k T_o B}_{N_i}$$

Noise temp. of input referred added noise.

$$N_a = N_o - G N_i \Rightarrow \frac{N_a}{G N_i} = F - 1$$

$$\Rightarrow F = \frac{N_a}{G N_i} + 1$$

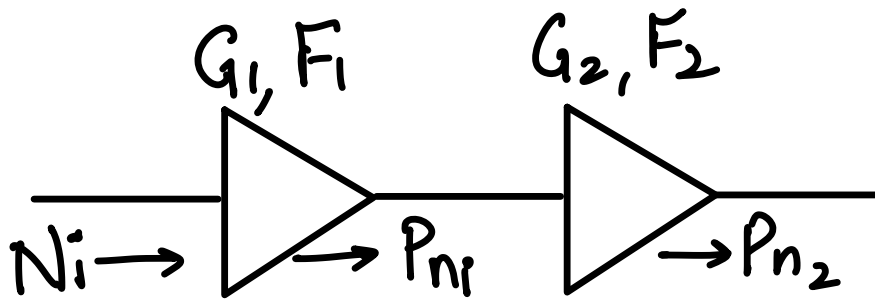
$$\Rightarrow F = 1 + \frac{k T_e B G}{G k T_o B}$$

$$\Rightarrow \boxed{F = 1 + \frac{T_e}{T_o}}$$

NF $\rightarrow$ Used for room temp.

$T_e \rightarrow$ Cryogenic

Noise Figure of Cascaded Systems



Assumptions

- 1) Noise of each stage is uncorrelated
- 2) F is independent of frequency.

$$P_{n1} = G_1 F_1 N_i$$

$$P_{n2} = G_2 P_{n1} + N_{a2}$$

$$P_{n2} = G_2 G_1 F_1 N_i + (F_2 - 1) G_2 N_i$$

$$F_{\text{total}} = \frac{P_{n2}}{G_1 G_2 N_i} = F_1 + \frac{F_2 - 1}{G_1}$$

$$F_{\text{total}} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \dots + \frac{F_N - 1}{G_1 G_2 \dots G_{N-1}}$$

Noise Measure

$$(F_1, G_1) \quad (F_2, G_2)$$

$$F_{12} = F_1 + \frac{F_2 - 1}{G_1}$$

$$F_{21} = F_2 + \frac{F_1 - 1}{G_2}$$

Say $F_{12} < F_{21}$

$$\Rightarrow F_1 + \frac{F_2 - 1}{G_1} < F_2 + \frac{F_1 - 1}{G_2}$$

$$\Rightarrow (F_1 - 1) + \frac{F_2 - 1}{G_1} < (F_2 - 1) + \frac{F_1 - 1}{G_2}$$

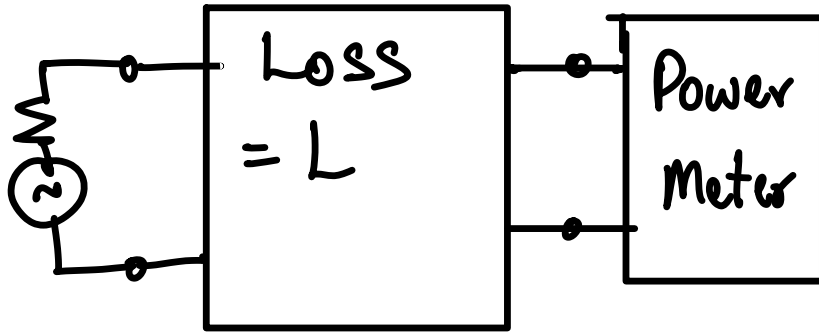
$$\Rightarrow (F_1 - 1) \left[1 - \frac{1}{G_2} \right] < (F_2 - 1) \left[1 - \frac{1}{G_1} \right]$$

$$\Rightarrow \frac{F_1 - 1}{1 - G_1^{-1}} < \frac{F_2 - 1}{1 - G_2^{-1}}$$

$M_N = \frac{F - 1}{1 - G^{-1}}$

 $\rightarrow$ Noise Measure

Noise Figure of an Attenuator



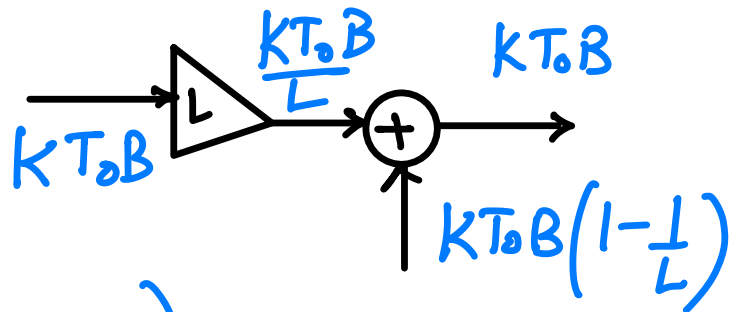
$$L = \frac{P_i}{P_o}$$

$$P_i = KT_o B$$

$$P_o = \frac{KT_o B}{L}$$

$$P_{PM} = KT_o B$$

Assume it is matched & at Thermal Equilibrium with env.



$$\Rightarrow N_a = KT_o B \left(1 - \frac{1}{L}\right)$$

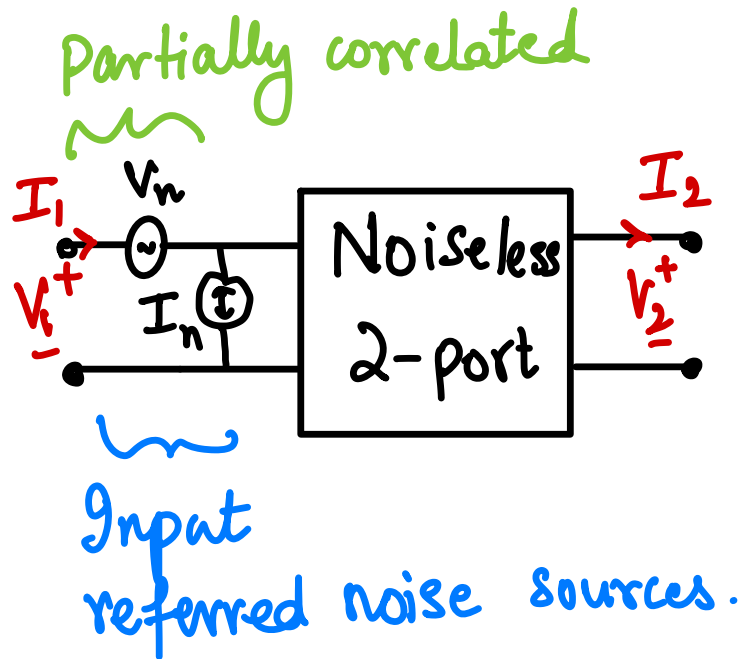
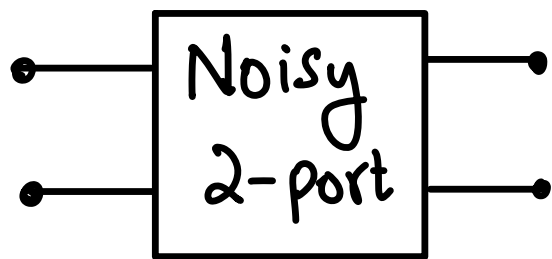
$$\Rightarrow F = 1 + \frac{N_a}{G N_i} = 1 + \frac{1}{KT_o B} \left(KT_o B \left(1 - \frac{1}{L}\right) \right)$$

$$= 1 + L - 1$$

$$\Rightarrow \boxed{F = L}$$

Noise Figure Dependence on Source Impedance

Noise Parameters



$$V_1 = AV_2 + BI_2 + V_n$$

$$I_1 = CV_2 + DI_2 + I_n$$

$$\overline{V_n^2} = 4kT_0 B R_N$$

$R_N \rightarrow$ y/p Referred Noise resistance

$$\overline{I_n^2} = 4kT_0 B G_N$$

$$I_n = \underbrace{Y_c V_n}_{\text{correlated}} + \underbrace{I_n}_{\text{uncorrelated}}$$

$Y_c = G_c + jB_c$ is the correlation admittance.

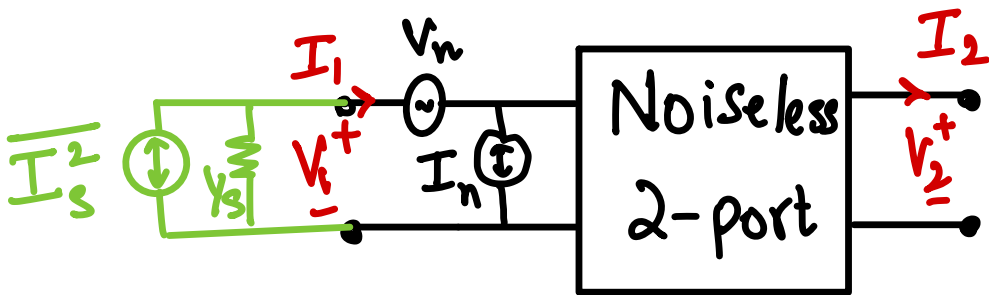
$$\overline{I_u^2} = 4 k T_0 B G_u$$

Input referred noise is fully determined by 4 noise parameters:

$R_N \rightarrow$ Noise resistance

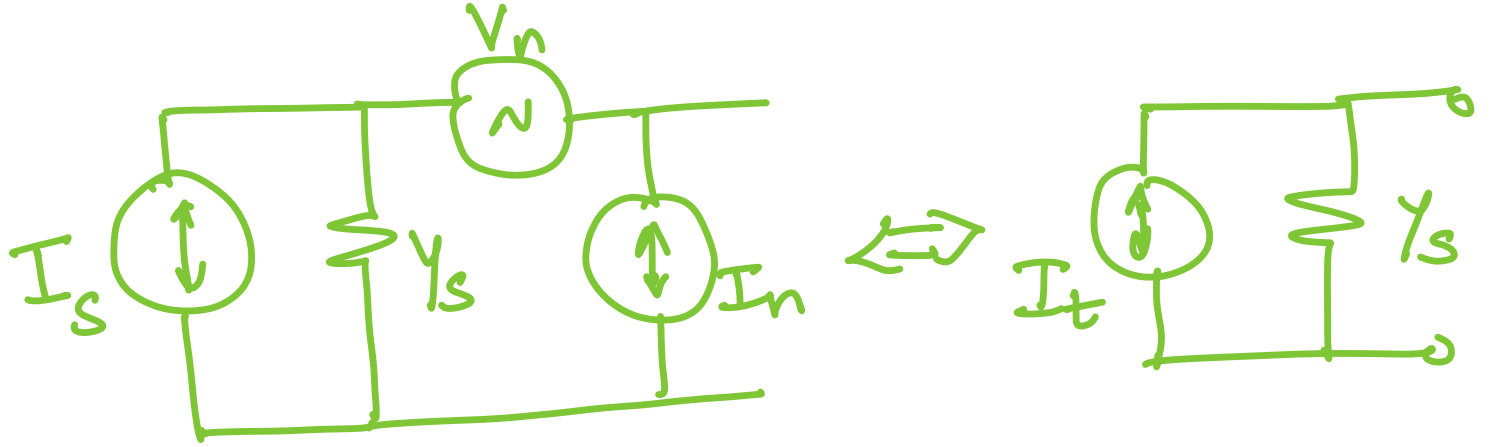
$G_u \rightarrow$ uncorrelated conductance

G_c
 B_c } Correlated cond. & susceptance.



$$\overline{I_s^2} = 4KT_0 B G_s \quad \text{where} \quad Y_s = G_s + jB_s$$

$$F = \frac{|I_t|^2 \rightarrow \text{total input referred noise}}{|I_s|^2 \rightarrow \text{Source noise}}$$



$$I_t = I_s + I_n + Y_s V_n$$

$$\Rightarrow F = \frac{|I_s|^2 + |I_n + Y_s V_n|^2}{|I_s|^2}$$

$$= 1 + \frac{|I_n + (Y_c + Y_s)V_n|^2}{|I_s|^2}$$

$$\Rightarrow F = 1 + \frac{|I_n|^2}{|I_s|^2} + |Y_c + Y_s|^2 \frac{|V_n|^2}{|I_s|^2}$$

$$F = 1 + \frac{G_u}{G_s} + \frac{R_N}{G_s} \left[(G_s + G_c)^2 + (B_s + B_c)^2 \right]$$

Noise Matching

$$B_s^{\text{opt}} = -B_c$$

$$G_s^{\text{opt}} \text{ from } \frac{dF}{dG_s} = 0$$

$$\Rightarrow G_s^{\text{opt}} = \sqrt{G_c^2 + \frac{G_u}{R_N}}$$

$$Y_s^{\text{opt}} = G_s^{\text{opt}} + j B_s^{\text{opt}}$$

$$F_{\min} = 1 + 2 R_N (G_c + G_s^{\text{opt}})$$

$$F = F_{\min} + \frac{R_N}{G_s} \left| Y_s - Y_s^{\text{opt}} \right|^2$$

Noise Parameters

$$F_{\min}, R_N, G_s^{\text{opt}}, B_s^{\text{opt}}.$$

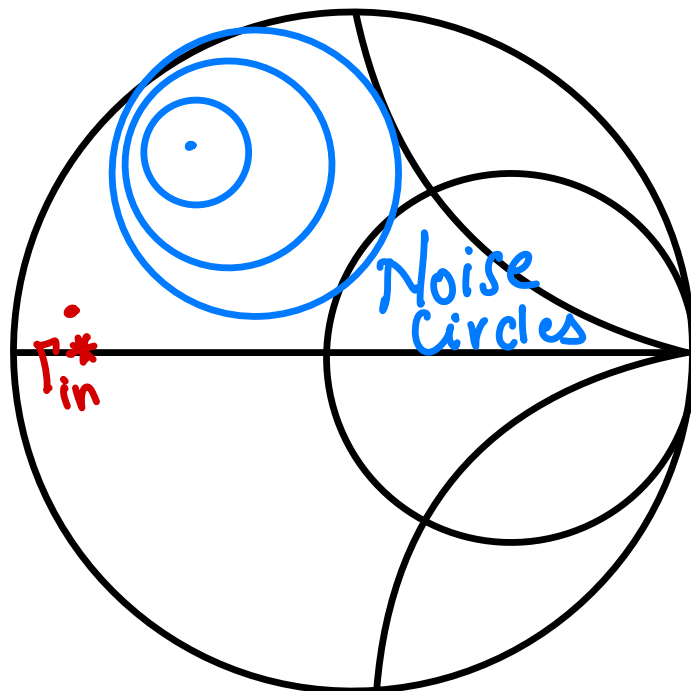
Noise Circles

$$Y_s = \frac{1}{Z_0} \frac{1 - \Gamma_s}{1 + \Gamma_s} \rightarrow \textcircled{1}$$

$$Y_s^{\text{opt}} = \frac{1}{Z_0} \frac{1 - \Gamma_s^{\text{opt}}}{1 + \Gamma_s^{\text{opt}}} \quad \Gamma_s^{\text{opt}} \rightarrow \textcircled{2}$$

$$N \triangleq \frac{F - F_{\min}}{4 R_N} |1 + \Gamma_s^{\text{opt}}|^2 \quad \textcircled{3}$$

$$\left| \Gamma_s - \underbrace{\frac{\Gamma_s^{\text{opt}}}{1+N}}_{\text{Center}} \right| = \underbrace{\frac{N(N+1 - |\Gamma_s^{\text{opt}}|^2)}{(1+N)^2}}_{\text{radius}^2}$$



Measuring Noise Parameters:
Source-Pull.